University of St.Gallen

Course and Examination Fact Sheet: Spring Semester 2026

8,037: Agentic AI Design, Governance and Management

ECTS credits: 6

Overview examination/s

(binding regulations see below)
decentral - Presentation, Analog, Group work individual grade (40%)
Examination time: Term time
decentral - Written work, Digital, Individual work individual grade (30%)
Examination time: Term time
decentral - Written work, Digital, Group work group grade (30%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
8,037,1.00 Agentic AI Design, Governance and Management -- English -- Ritz Eva , Li Mahei

Course information

Prerequisites

Interest for AI within organizational contexts

Learning objectives

- You can understand the conceptual foundations of agentic AI, including reasoning, tool use, and goal-directed behavior.
- You can explain differences between LLMs, agentic systems, and classical automation architectures.
- You can explain and apply methods to structure a service design challenge.
- Describe and design value creation processes in service systems from a value co-creation perspective.
- You can describe and apply the principles and methods of user-centric design and analyze user journeys, pain points, and service touchpoints.
- You can explain and apply agile methods for service design.
- You can explain the role of business processes and workflows, you can model those and design agentic architectures in the context of agentic AI.
- You can explain design patterns for domain-based agents, and apply them when building agents.
- You can explain, analyze, and create context engineering strategies for domain-based agents.
- You can explain the fundamentals of multi-agent-systems, agent coordination and orchestration between agents and analyze interactions and dependencies between agents.
- You can apply multi-agent design patterns and create a multi-agent prototype for a specific workflow.
- You can explain risks associated with agentic AI and identify suitable mitigation strategies.
- You can explain and analyze governance mechanisms cross deployment scenarios and create a risk-aware governance plan for an multi-agent prototype.
- You can analyze how agentic systems redistribute cognitive labor, autonomy, and decision rights.
- You can explain the logic of change management, and digital transformation and apply it toward agentic projects

Content

The course *Agentic AI: Foundations, Architectures, and Governance* examines the theoretical, methodological, and applied dimensions of agentic artificial intelligence in contemporary organizational contexts. The course enables students to develop a comprehensive understanding of how agentic AI reshapes work processes, service delivery, and digital transformation. It is conducted in close collaboration with **IBM**, providing students with access to cutting-edge technological frameworks, enterprise tools, hands-on experiences, and real-world design challenges.

Students learn to structure innovation challenges using service design methods and fundamentals of service science, analyze user needs and service touchpoints, and apply user-centric and agile design approaches to agentic solution development. Building on this, the course introduces techniques for modeling business processes and workflows as a basis for designing domain-specific agent architectures. Students apply design patterns for domain-based agents and develop context engineering strategies that enable robust agent performance.

A further part focuses on multi-agent systems. Students study coordination and orchestration mechanisms, analyze interactions and dependencies across agents, and develop a functioning multi-agent prototype for a selected workflow.

The course concludes with an overview of risk management and governance of agentic AI. Students identify risks across deployment scenarios, analyze governance mechanisms, and design a risk-aware governance plan for their multi-agent prototype.

Through the integration of conceptual insights, methodological frameworks, and hands-on collaboration with IBM technologies, the course equips students with the competencies required to design, implement, and govern agentic AI systems in organizational settings.

Structure and indications of the learning and teaching design

- This course counts 6 credits. Accordingly, the total workload for students is 180 hours. This includes self-study, campus time and all grading-relevant extra actvities, such as group work and the written synopsis.
- The structure of the contact hours is as follows: 48h hours of weekly lectures during the semester, 4h each week (excluding break).
- The course is conducted in presence in St. Gallen.

The course structure is as following:

1. Agentic AI Foundations
2. Service Design and Service Systems Foundations
3. Service Design Methods
4. Agentic AI Workflows
5. Designing AI Agents
6. Context Engineering for AI Agents
7. Multi-Agent System Foundations
8. Build your own Multi-Agent System
9. Managing Risks of AI Agents
10. AI Agents and the future of work

In parallel to the lecture, students apply the methods they have learned as part of a running case. In addition to developing their own solutions and use existing tools (e.g. Langflow and IBM watsonx) to build their own agents. This aims to foster action-oriented and transfer competencies.

Literature

Jan Marco Leimeister (2020). Dienstleistungsengineering und -management

Jan Marco Leimeister (2025). Einführung in die Wirtschaftsinformatik

Pan et al. (2025). Measuring Agents in Production

Additional information

This course is happening in collaboration with IBM Switzerland.

In case there are any questions, please contact:

Dr. Mahei Li, Institute of Information Systems and Digital Business, mahei.li@unisg.ch

Dr. Eva Ritz, IBM Switzerland, eva.ritz@unisg.ch

Examination information

Examination sub part/s

1. Examination sub part (1/3)

Examination modalities

Examination type	Presentation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work individual grade
Weighting	40%
Duration	--

Languages

Question language: English
Answer language: English

Remark

Group presentation of running project

Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

The use of Generative AI tools (e.g., ChatGPT, Copilot, Gemini) is permitted as a supportive instrument in this course (e.g., for idea generation, structuring, language refinement, or clarification of concepts). Students remain fully responsible for all submitted work. The use of Generative AI must not replace independent analysis, conceptual reasoning, or critical reflection. Content generated using generative AI systems must be clearly labeled as such.

Presentation

- Generative AI may be used to support slide structuring, wording, or visualization ideas.
- The presentation content and oral explanations must reflect the students' own understanding.
- Students must be able to explain and defend all presented content independently.

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2. Examination sub part (2/3)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous

Examination location	Off Campus
Grading type	Individual work individual grade
Weighting	30%
Duration	--

Languages
Question language: English
Answer language: English

Remark
Individual synopsis after group presentations

Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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Individual Synopsis

- Generative AI may be used for language editing or structural support.
- The analytical content, synthesis, and conclusions must be independently developed.
- Submitting AI-generated text without substantial own intellectual contribution constitutes a violation of academic integrity.

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3. Examination sub part (3/3)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	Off Campus
Grading type	Group work group grade
Weighting	30%
Duration	--

Languages
Question language: English
Answer language: English

Remark
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Examination-aid rule
Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

The use of Generative AI tools (e.g., ChatGPT, Copilot, Gemini) is permitted as a supportive instrument in this course (e.g., for idea generation, structuring, language refinement, or clarification of concepts). Students remain fully responsible for all submitted work. The use of Generative AI must not replace independent analysis, conceptual reasoning, or critical reflection. Content generated using generative AI systems must be clearly labeled as such.

Group Work

- Generative AI may be used for brainstorming, outlining, or refining group deliverables.
- All group members must demonstrate shared understanding and authorship of the submitted work.
- AI-generated content must be critically reviewed and adapted by the group.

Content

All content covered in the lectures, all lecture materials, and the examination literature are relevant for the exam. This also includes any guest lectures. The examination content is derived from the learning objectives of the course. The exam assessment consists of group work, a presentation, and a final individual assignment (synopsis).

Literature

All materials made available on Canvas up to the end of the course sessions, inc.l particular academic journal articles and white papers.

For further reading, please refer to the textbook: Leimeister, Jan Marco (2020): Dienstleistungsengineering und-management: Data-driven Service Innovation, Springer, Berlin.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW21) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 13 (Monday, 23rd March 2026) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 04 (Thursday, 22nd January 2026);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 12 (Monday, 16 March 2026);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 14 (Monday, 30 March 2026);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 15 (Monday, 06 April 2026).